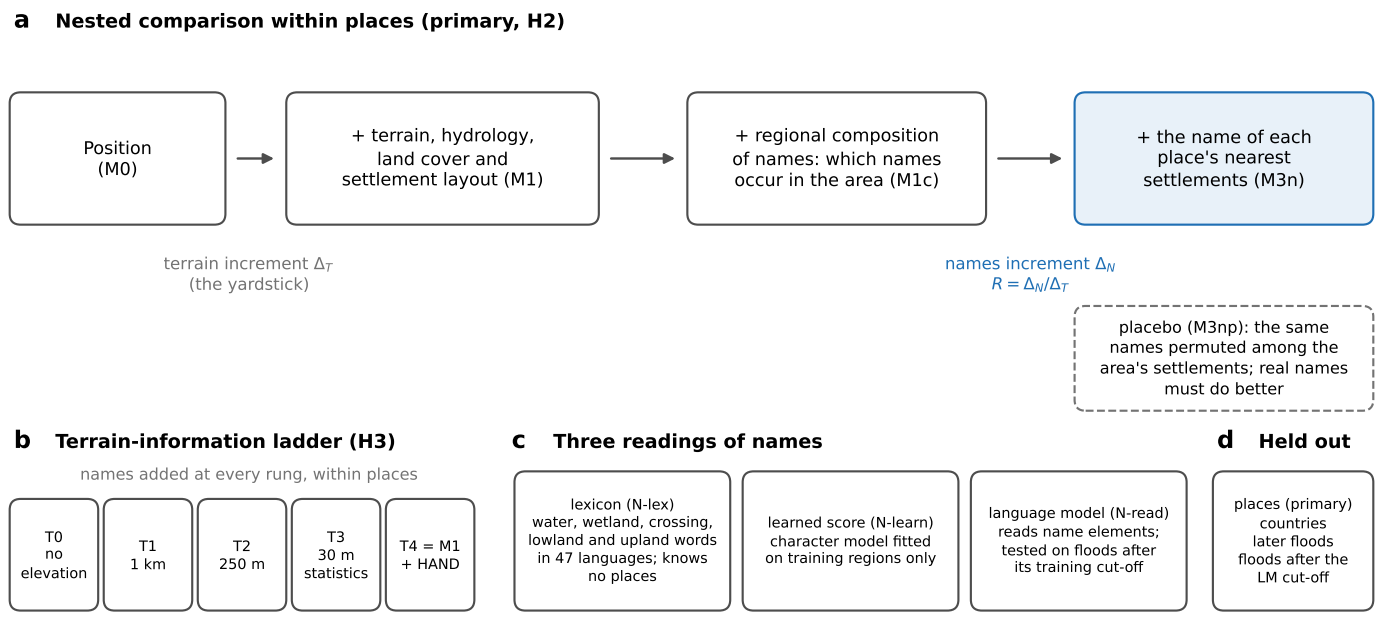
a
Nested comparison within places (primary, H2)
Position (M0)
+ terrain, hydrology, land cover and settlement layout (M1)
+ regional composition of names: which names occur in the area (M1c)
+ the name of each place's nearest settlements (M3n)
terrain increment Δ_T (the yardstick)
names increment Δ_N
$R = \Delta_N/\Delta_T$
placebo (M3np): the same names permuted among the area's settlements; real names must do better
b
Terrain-information ladder (H3)
names added at every rung, within places
T0 no elevation
T1 1 km
T2 250 m
T3 30 m statistics
T4 = M1 + HAND
c
Three readings of names
lexicon (N-lex) water, wetland, crossing, lowland and upland words in 47 languages; knows no places
learned score (N-learn) character model fitted on training regions only
language model (N-read) reads name elements; tested on floods after its training cut-off
d
Held out
places (primary)
countries
later floods
floods after the LM cut-off